

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 3.1

Addition and Multiplication of Two Sequences

Aim:

To perform and verify the addition and multiplication of two discrete-time sequences using Scilab. The Aim is to understand how basic arithmetic operations on sequences work and to observe the resulting sequences after performing addition and multiplication

Program:

% Test Case: Addition and Multiplication of Two Sequences

clc;

clear;

close all;

% Define parameters

N = 30; % Number of samples

n = 0:N-1; % Discrete time index vector

% Define Sequence 1: Sinusoidal sequence

f1 = 0.05; % Frequency in cycles/sample

A1 = 2; % Amplitude

x1_n = A1 * sin(2 * pi * f1 * n);

```
% Define Sequence 2: Exponentially decaying sequence
```

```
a2 = 0.1;          % Decay constant
```

```
A2 = 3;           % Amplitude
```

```
x2_n = A2 * exp(-a2 * n);
```

```
% Addition of the two sequences
```

```
x_add = x1_n + x2_n;
```

```
% Multiplication of the two sequences
```

```
x_mult = x1_n .* x2_n;
```

```
% Plot the sequences and their results
```

```
figure;
```

```
% Plot Sequence 1
```

```
subplot(4,1,1);
```

```
stem(n, x1_n, 'filled');
```

```
grid on;
```

```
xlabel('n (Discrete Time Index)');
```

```
ylabel('Amplitude');
```

```
title('Sequence 1: Sinusoidal Sequence');
```

```
% Plot Sequence 2
```

```
subplot(4,1,2);
```

```
stem(n, x2_n, 'filled');
```

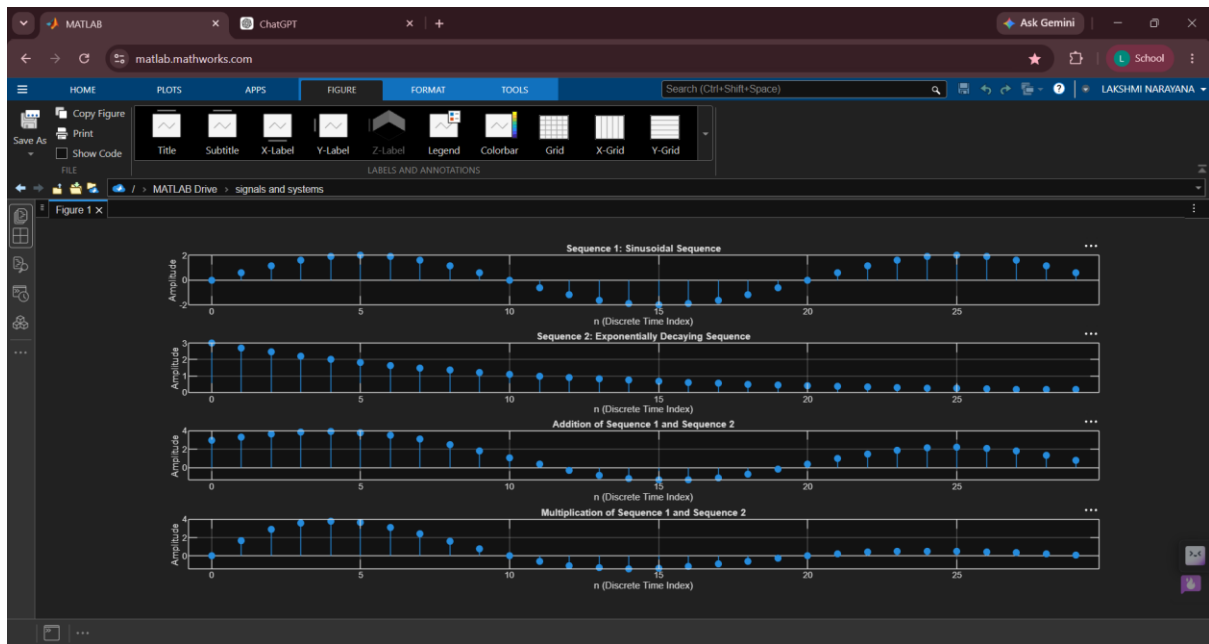
```
grid on;
```

```
xlabel('n (Discrete Time Index)');  
ylabel('Amplitude');  
title('Sequence 2: Exponentially Decaying Sequence');
```

```
% Plot the result of addition  
subplot(4,1,3);  
stem(n, x_add, 'filled');  
grid on;  
xlabel('n (Discrete Time Index)');  
ylabel('Amplitude');  
title('Addition of Sequence 1 and Sequence 2');
```

```
% Plot the result of multiplication  
subplot(4,1,4);  
stem(n, x_mult, 'filled');  
grid on;  
xlabel('n (Discrete Time Index)');  
ylabel('Amplitude');  
title('Multiplication of Sequence 1 and Sequence 2');
```

Output Waveform:



Result:

Addition: The addition of the two sequences combines their characteristics, leading to a sequence that reflects the contributions of both the sinusoidal and exponential components.

- **Multiplication:** The multiplication of the two sequences results in a sequence that has the characteristics of both, where the sinusoidal sequence is modulated by the exponential decay.

This test case demonstrates the process of adding and multiplying two sequences in the discrete-time domain, providing insight into how these operations affect the resulting signals.